Sao Aphisith Sithisack

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EDUCATION

University of Oklahoma **Norman, OK**
B.S. in Electrical & Computer Engineering (Honors) | Minor in Mathematics & Computer Science | GPA: 3.88 *Aug 2024 – May 2028*
Honors: Davis UWC Scholar (full tuition, 4 yrs) | Robert Vinton Clapp Scholarship

Relevant Coursework: Introduction to Electronics, Microcontrollers, Circuits Laboratory, Electromagnetic Fields, Differential Equations, Linear Algebra

EXPERIENCE

Pulsed Laser Template Engineering (PLATEN) Lab **Jan 2026 – Present**
Undergraduate Researcher — Semiconductor Nanofabrication *Norman, OK*

- Fabricate high-aspect-ratio micro- and nanostructures in a cleanroom using photolithography, reactive ion etching (RIE), and pulsed laser deposition (PLD).
- Independently own the PLD step end to end, growing LAO films on STO substrates and monitoring growth with RHEED.
- Optimize anisotropic plasma etch parameters (gas chemistry, RF power, chamber pressure) and mask selectivity to reach vertical sidewalls and controlled feature depth.

Sooner Competitive Robotics **Jan 2026 – Present**
Electrical Systems Engineer — Power Distribution *Norman, OK*

- Design a **4-layer** power, CAN, and E-Stop distribution PCB in **Altium** for the IGVC 2027 robot, routing 12 V at up to **10 A** to **7 board ports**, each with INA220 current sensing (layout in progress).
- Replaced the team-standard Schottky with P-MOSFET reverse-polarity protection, cutting calculated loss at 8 A from **3.6 W to 0.7 W**.
- Specified the 3.3 V synchronous buck converter and the 8 MHz crystal clock for the 125 kbps CAN bus, recording each design decision for team and captain review.

Student Learning Center **Aug 2025 – Present**
Peer Learning Assistant | 20 hrs/week *Norman, OK*

- Mentor **250+ students per semester** in Physics I–II, Calculus I–III, Differential Equations, Linear Algebra, and General Chemistry I through structured problem-solving.

Light–Matter Interaction Lab (Dr. Tischler) **Aug 2025 – Dec 2025**
Undergraduate Researcher — Optical Characterization *Norman, OK*

- Characterized silicon carbide nanocrystals by **Raman spectroscopy**, aligning the benchtop optical setup and collecting spectra to assess crystal structure and material quality.

Cloud Microphysics Group, First Year Research Experience **Jan 2025 – May 2025**
Undergraduate Researcher — Data Analysis *Norman, OK*

- Processed NSF/NCAR C-130 aircraft data from the CAESAR campaign in **Python**, comparing cloud water content, particle size, and condensation-nuclei concentration across probes versus altitude; presented the results as a research poster.

PROJECTS

TI MSPM0 USB-C Microcontroller Board **Sep 2026 – Present**
KiCad | 2-layer PCB | write-up *Personal Project*

- Designed the schematic for a USB-C powered **Texas Instruments MSPM0G3507** board: 3.3 V LDO, CH340E USB-to-UART bridge, USBLC6 ESD protection, 8 MHz crystal, and SWD debug; routing the 2-layer layout.
- Set design targets from USB 2.0 limits and datasheets: $\leq$100 mA load, $\leq$0.2 W LDO loss, ADC reference noise $<$1 LSB.

LEADERSHIP

United World College Lao PDR National Committee **Jul 2024 – Present**
Co-President & Alumni Coordinator *Remote / Laos*

- Co-lead a **20+ volunteer** committee running Laos' UWC scholarship cycle, managing **150+ applicants** per cycle down to 3–4 placements, and led fundraising exceeding **$2,500**.

Association of Southeast Asian Nations at OU **Nov 2024 – May 2026**
Sponsorship & Volunteering Chair *Norman, OK*

- Secured $1,000+ in external funding through direct outreach to 20+ donors and organized cultural events.

Activities: IEEE Student Branch at OU, Event Committee | Sooner Racing Team, Electrical (Sep 2026 – Present)

TECHNICAL SKILLS

PCB & Hardware: Altium Designer (4-layer PCB), KiCad (schematic capture), power protection and distribution, USB-C, UART, CAN, I^2C, SWD
Fabrication & Lab: Photolithography, anisotropic RIE, pulsed laser deposition, RHEED, cleanroom processing, Raman spectroscopy, oscilloscope, DMM, function generator, bench supply
Programming: C, C++, Python, MATLAB, LabVIEW, Linux, Git